Complex Analysis Homework 2

1. Let n be a positive integer.

(a) (i) Find all the roots of $z^{2n}+1=0$.

(ii) By factorizing $z^{2n}+1$ into a product of quadratic factors with real coefficients, prove that

$$z^n+\frac{1}{z^n}=\prod_{k=0}^{n-1}\left(z+\frac{1}{z}-2\cos\frac{(2k+1)\pi}{2n}\right) \text{ for all } z\neq 0 \text{ and}$$

$$\cos n\theta=2^{n-1}\prod_{k=0}^{n-1}\left(\cos\theta-\cos\frac{(2k+1)\pi}{2n}\right)$$

(iii) Using (a)(ii), prove that $\prod_{k=0}^{n-1}\cos\frac{(2k+1)\pi}{4n}=\prod_{k=0}^{n-1}\sin\frac{(2k+1)\pi}{4n}=\frac{\sqrt{2}}{2^n}$.

(b) Let $n>1$.

(i) Find all the roots of $z^{2n}-1=0$.

(ii) By factorizing $z^{2n}-1$ into a product of quadratic factors with real coefficients, prove that

$$z^n-\frac{1}{z^n}=\left(z-\frac{1}{z}\right)\prod_{k=1}^{n-1}\left(z-2\cos\frac{k}{n}\pi+\frac{1}{z}\right) \text{ for all } z\neq 0 \text{ and}$$

$$\sin n\theta=2^{n-1}\sin\theta\prod_{k=1}^{n-1}\left(\cos\theta-\cos\frac{k\pi}{n}\right)$$

(iii) Using (b)(ii), prove that $\prod_{k=1}^{n-1}\sin\frac{k\pi}{2n}=\prod_{k=1}^{n-1}\cos\frac{k\pi}{2n}=\frac{\sqrt{n}}{2^{n-1}}$.

Solution

1. (a) (i) $z^{2n}+1=0$

$$z^{2n}=-1$$

$$z^{2n}=\cos\pi+i\sin\pi$$

$$z=\cos\frac{(2k+1)\pi}{2n}+i\sin\frac{(2k+1)\pi}{2n},\ \text{where}\ k=0,1,2,\cdots,2n-1\ .$$

(ii) Let $z_k=\cos\frac{(2k+1)\pi}{2n}+i\sin\frac{(2k+1)\pi}{2n}$ for all $k=0,1,2,\cdots,2n-1$.

Then, $z_{2n-k-1}=\cos\frac{\big(2(2n-k-1)+1\big)\pi}{2n}+i\sin\frac{\big(2(2n-k-1)+1\big)\pi}{2n}$

$$=\cos\left(2\pi-(\frac{2k+1}{2n})\pi\right)+i\sin\left(2\pi-(\frac{2k+1}{2n})\pi\right)$$

$$=\cos\frac{(2k+1)\pi}{2n}-i\sin\frac{(2k+1)\pi}{2n}$$

$$=\overline{z_k}\ \text{ for all }\ k=0,1,2,\cdots,n-1$$

$$\therefore\quad z^{2n}+1=\prod_{k=0}^{2n-1}(z-z_k)$$

$$=\prod_{k=0}^{n-1}\Big((z-z_k)(z-\overline{z_k})\Big)$$

$$=\prod_{k=0}^{n-1}\Big(z^2-(z_k+\overline{z_k})z+z_k\overline{z_k}\Big)$$

$$=\prod_{k=0}^{n-1}\left(z^2-2z\cos\frac{(2k+1)\pi}{2n}+1\right)$$

Thus, we have

$$\frac{1}{z^n}(z^{2n}+1)=\frac{1}{z^n}\prod_{k=0}^{n-1}\left(z^2-2z\cos\frac{(2k+1)\pi}{2n}+1\right)\qquad\text{for all}\ z\neq 0$$

$$z^n+\frac{1}{z^n}=\prod_{k=0}^{n-1}\left(z-2\cos\frac{(2k+1)\pi}{2n}+\frac{1}{z}\right)\qquad\text{for all}\ z\neq 0$$

$$z^n+\frac{1}{z^n}=\prod_{k=0}^{n-1}\left(z+\frac{1}{z}-2\cos\frac{(2k+1)\pi}{2n}\right)\qquad\text{for all}\ z\neq 0$$

Let $z=\cos\theta+i\sin\theta$. We have

$$2\cos n\theta = z^n + \frac{1}{z^n}$$
$$= \prod_{k=0}^{n-1}\left(z + \frac{1}{z} - 2\cos\frac{(2k+1)\pi}{2n} \right)$$
$$= \prod_{k=0}^{n-1}\left(2\cos\theta - 2\cos\frac{(2k+1)\pi}{2n} \right)$$
$$= 2^n \prod_{k=0}^{n-1}\left(\cos\theta - \cos\frac{(2k+1)\pi}{2n} \right)$$
$$\cos n\theta = 2^{n-1} \prod_{k=0}^{n-1}\left(\cos\theta - \cos\frac{(2k+1)\pi}{2n} \right)$$

(b) Put $z = -1$ into (a)(ii), we have

$$2(-1)^n = \prod_{k=0}^{n-1}\left(-2 - 2\cos\frac{(2k+1)\pi}{2n} \right)$$
$$2(-1)^n = \prod_{k=0}^{n-1}\left((-2)\left(2\cos^2(\frac{(2k+1)\pi}{4n}) \right) \right)$$
$$2(-1)^n = (-2)^n (2^n) \prod_{k=0}^{n-1} \cos^2\left(\frac{(2k+1)\pi}{4n} \right)$$
$$\prod_{k=0}^{n-1} \cos^2\left(\frac{(2k+1)\pi}{4n} \right) = \frac{2}{2^{2n}}$$

Note that $0 < \frac{(2k+1)\pi}{4n} < \frac{\pi}{2}$ for all $k = 0, 1, 2, \cdots, n-1$, we have

$\cos\frac{(2k+1)\pi}{4n} > 0$ for all $k = 0, 1, 2, \cdots, n-1$.

So, $\prod_{k=0}^{n-1} \cos\frac{(2k+1)\pi}{4n} > 0$

Thus, $\prod_{k=0}^{n-1} \cos\frac{(2k+1)\pi}{4n} = \frac{\sqrt{2}}{2^n}$

$$\prod_{k=0}^{n-1} \sin\frac{(2k+1)\pi}{4n} = \prod_{k=0}^{n-1} \cos\left(\frac{\pi}{2} - \frac{(2k+1)\pi}{4n} \right) = \prod_{k=0}^{n-1} \cos\left(\frac{(2n-2k-1)\pi}{4n} \right) = \prod_{k=0}^{n-1} \cos\frac{(2k+1)\pi}{4n} = \frac{\sqrt{2}}{2^n}.$$

(b)(i) $z^{2n}-1=0$

$z^{2n}=1=\cos 0+i\sin 0$

$z=\cos\dfrac{2k\pi}{2n}+i\sin\dfrac{2k\pi}{2n}=\cos\dfrac{k\pi}{n}+i\sin\dfrac{k\pi}{n},\quad$ where $k=0,1,\cdots,2n-1$

(ii) Let $z_k=\cos\dfrac{k\pi}{n}+i\sin\dfrac{k\pi}{n},$ where $k=0,1,\cdots,2n-1.$ Then

$$\begin{aligned}
z_{2n-k}&=\cos\frac{(2n-k)\pi}{n}+i\sin\frac{(2n-k)\pi}{n}\\
&=\cos\left(2\pi-\frac{k\pi}{n}\right)+i\sin\left(2\pi-\frac{k\pi}{n}\right)\\
&=\cos\frac{k\pi}{n}-i\sin\frac{k\pi}{n}\\
&=\overline{z_k}\quad \text{for } k=1,\cdots,n-1
\end{aligned}$$

$$\begin{aligned}
z^{2n}-1&=\prod_{k=0}^{2n-1}(z-z_k)\\
&=(z-1)(z+1)\prod_{k=1}^{n-1}\left((z-z_k)(z-\overline{z_k})\right)\\
&=(z^2-1)\prod_{k=1}^{n-1}\left(z^2-2z\cos\frac{k\pi}{n}+1\right)\quad \text{for all } z\neq 0
\end{aligned}$$

$$z^n-\frac{1}{z^n}=\left(z-\frac{1}{z}\right)\prod_{k=1}^{n-1}\left(z-2\cos\frac{k\pi}{n}+\frac{1}{z}\right)\quad \text{for all } z\neq 0$$

Let $z=\cos\theta+i\sin\theta.$ We have

$$\begin{aligned}
2\sin n\theta&=z^n-\frac{1}{z^n}\\
&=\left(z-\frac{1}{z}\right)\prod_{k=1}^{n-1}\left(z-2\cos\frac{k\pi}{n}+\frac{1}{z}\right)\\
&=2\sin\theta\prod_{k=1}^{n-1}\left(2\cos\theta-2\cos\frac{k\pi}{n}\right)\\
&=2^n\sin\theta\prod_{k=1}^{n-1}\left(\cos\theta-\cos\frac{k\pi}{n}\right)\\
\sin n\theta&=2^{n-1}\sin\theta\prod_{k=1}^{n-1}\left(\cos\theta-\cos\frac{k\pi}{n}\right)
\end{aligned}$$

(iii) $\lim_{\theta\to 0}\frac{\sin n\theta}{\sin\theta}=2^{n-1}\lim_{\theta\to 0}\prod_{k=1}^{n-1}\left(\cos\theta-\cos\frac{k\pi}{n}\right)$

$$n=2^{n-1}\prod_{k=1}^{n-1}\left(1-\cos\frac{k\pi}{n}\right)=2^{n-1}\prod_{k=1}^{n-1}\left(2\sin^2\frac{k\pi}{2n}\right)=2^{2n-2}\prod_{k=1}^{n-1}\left(\sin^2\frac{k\pi}{2n}\right)$$

$$\prod_{k=1}^{n-1}\left(\sin^2\frac{k\pi}{2n}\right)=\frac{n}{2^{2n-2}}$$

$$\prod_{k=1}^{n-1}\left(\sin\frac{k\pi}{2n}\right)=\frac{\sqrt{n}}{2^{n-1}}\quad\left[\because \sin\frac{k\pi}{2n}>0\quad \text{for } k=1,\cdots,n-1\right]$$

$$\prod_{k=1}^{n-1}\left(\cos\frac{k\pi}{2n}\right)=\prod_{k=1}^{n-1}\left(\sin\left(\frac{\pi}{2}-\frac{k\pi}{2n}\right)\right)=\prod_{k=1}^{n-1}\left(\sin\frac{(n-k)\pi}{2n}\right)=\prod_{k=1}^{n-1}\left(\sin\frac{k\pi}{2n}\right)=\frac{\sqrt{n}}{2^{n-1}}$$

2. Suppose $f(z)=u(z)+iv(z)$ is an analytic function and the polar coordinate of z is (r, θ).

(a) Using

(i) Method 1: chain rule

(ii) Method 2: the difference quotient $\frac{f(z+\Delta z)-f(z)}{\Delta z}$, as $z+\Delta z\to z=re^{i\theta}$ along the ray $\arg z=\theta$ and along the circle $|z|=r$,

show that

$$\frac{\partial u}{\partial r}=\frac{1}{r}\frac{\partial v}{\partial\theta},\quad \frac{\partial v}{\partial r}=-\frac{1}{r}\frac{\partial u}{\partial\theta}.$$

(b) Determine the set on which

$$f(z)=\begin{cases} z & \text{if } |z|\le 1\\ z^2 & \text{if } |z|>1\end{cases}$$

is analytic and compute its derivative.

Solution

(ai) $z = re^{i\theta} \Longrightarrow x = r\cos\theta$ and $y = r\sin\theta$ and

$$f(z) = u(x(r,\theta), y(r,\theta)) + iv(x(r,\theta), y(r,\theta))$$

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta$$

Similar applications of the chain rule yield

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}(-r\sin\theta) + \frac{\partial u}{\partial y}r\cos\theta$$
$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta$$
$$\frac{\partial v}{\partial \theta} = \frac{\partial v}{\partial x}(-r\sin\theta) + \frac{\partial v}{\partial y}r\cos\theta$$

Replace the partial derivatives on the right sides of the equations for $\frac{\partial u}{\partial r}$ and $\frac{\partial v}{\partial r}$ by their Cauchy-Riemann counterparts to obtain:

$$\frac{\partial u}{\partial r} = \frac{\partial v}{\partial y}\cos\theta - \frac{\partial v}{\partial x}\sin\theta = \frac{1}{r}\frac{\partial v}{\partial \theta}$$
$$\frac{\partial v}{\partial r} = -\frac{\partial u}{\partial y}\cos\theta + \frac{\partial u}{\partial x}\sin\theta = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

(ii) case (1): fix r, $\Delta\theta \to 0$. $z + \Delta z = re^{i(\theta+\Delta\theta)} \to z = re^{i\theta}$

$$\lim_{\Delta z\to 0}\frac{f(z+\Delta z)-f(z)}{\Delta z} = \lim_{\Delta\theta\to 0}\frac{f(re^{i(\theta+\Delta\theta)})-f(re^{i\theta})}{re^{i(\theta+\Delta\theta)}-re^{i\theta}}$$
$$= \frac{1}{re^{i\theta}}\lim_{\Delta\theta\to 0}\frac{f(re^{i(\theta+\Delta\theta)})-f(re^{i\theta})}{\Delta\theta}\frac{\Delta\theta}{e^{i\Delta\theta}-1}$$
$$= \frac{1}{re^{i\theta}}\lim_{\Delta\theta\to 0}\frac{u(re^{i(\theta+\Delta\theta)})-u(re^{i\theta})+i\left(v(re^{i(\theta+\Delta\theta)})-v(re^{i\theta})\right)}{\Delta\theta}\lim_{\Delta\theta\to 0}\frac{\Delta\theta}{e^{i\Delta\theta}-1}$$
$$= \frac{1}{ire^{i\theta}}\left(\frac{\partial u}{\partial\theta} + i\frac{\partial v}{\partial\theta}\right)$$

case (2): fix r, θ, $\Delta r \to 0$. $z + \Delta z = (r+\Delta r)e^{i\theta} \to z = re^{i\theta}$

$$\lim_{\Delta z\to 0}\frac{f(z+\Delta z)-f(z)}{\Delta z}=\lim_{\Delta r\to 0}\frac{f((r+\Delta r)e^{i\theta})-f(re^{i\theta})}{(r+\Delta r)e^{i\theta}-re^{i\theta}}$$

$$=\lim_{\Delta r\to 0}\frac{f((r+\Delta r)e^{i\theta})-f(re^{i\theta})}{\Delta re^{i\theta}}$$

$$=\frac{1}{e^{i\theta}}\lim_{\Delta r\to 0}\frac{u((r+\Delta r)e^{i\theta})-u(re^{i\theta})+i\left(v((r+\Delta r)e^{i\theta})-v(re^{i\theta})\right)}{\Delta r}$$

$$=\frac{1}{e^{i\theta}}\left(\frac{\partial u}{\partial r}+i\frac{\partial v}{\partial r}\right)$$

Since $f(z)=u(z)+iv(z)$ is an analytic function, we have

$$\frac{1}{ire^{i\theta}}\left(\frac{\partial u}{\partial \theta}+i\frac{\partial v}{\partial \theta}\right)=\frac{1}{e^{i\theta}}\left(\frac{\partial u}{\partial r}+i\frac{\partial v}{\partial r}\right)$$

$$\frac{\partial u}{\partial r}=\frac{1}{r}\frac{\partial v}{\partial \theta},\quad \frac{\partial v}{\partial r}=-\frac{1}{r}\frac{\partial u}{\partial \theta}$$

(b) $u=\begin{cases} r\cos\theta & 0\le r\le 1 \\ r^2\cos 2\theta & r>1 \end{cases}$ $v=\begin{cases} r\sin\theta & 0\le r\le 1 \\ r^2\sin 2\theta & r>1 \end{cases}$

$\frac{\partial u}{\partial r}=\begin{cases} \cos\theta & 0\le r<1 \\ 2r\cos 2\theta & r>1 \end{cases}$ $\frac{\partial v}{\partial r}=\begin{cases} \sin\theta & 0\le r<1 \\ 2r\sin 2\theta & r>1 \end{cases}$

$\frac{\partial u}{\partial \theta}=\begin{cases} -r\sin\theta & 0\le r<1 \\ -2r^2\sin 2\theta & r>1 \end{cases}$ $\frac{\partial v}{\partial \theta}=\begin{cases} r\cos\theta & 0\le r<1 \\ 2r^2\cos 2\theta & r>1 \end{cases}$

$\frac{\partial u}{\partial r}=\frac{1}{r}\frac{\partial v}{\partial \theta},\quad \frac{\partial v}{\partial r}=-\frac{1}{r}\frac{\partial u}{\partial \theta}$ everywhere except unit circle

$f'(z)=\begin{cases} 1 & \text{if } |z|<1 \\ 2z & \text{if } |z|>1 \end{cases}$

f is not analytic on the unit circle.

case 1: $z+\Delta z=e^{i(\theta+\Delta\theta)}\to z=e^{i\theta}$

$$\lim_{\Delta z\to 0}\frac{f(z+\Delta z)-f(z)}{\Delta z}=\frac{1}{ie^{i\theta}}\left(\frac{\partial u}{\partial \theta}+i\frac{\partial v}{\partial \theta}\right)=\frac{1}{ie^{i\theta}}\left(-\sin\theta+i\cos\theta\right)=1$$

case 2: $z+\Delta z=(1+\Delta r)e^{i\theta}\to z=e^{i\theta},\ \Delta r>0$

$$\lim_{\Delta z\to 0}\frac{f(z+\Delta z)-f(z)}{\Delta z}=\frac{1}{e^{i\theta}}\left(\frac{\partial u}{\partial r}+i\frac{\partial v}{\partial r}\right)=\frac{1}{e^{i\theta}}\left(2\cos 2\theta+2i\sin 2\theta\right)=2e^{i\theta}$$

3. Show that the function $f(z) = \sqrt{|\mathrm{Im}(z^2)|}$ satisfies the Cauchy–Riemann equations at $z = 0$, but is not differentiable at $z = 0$.

Solution

$$u(x, y) = \sqrt{|\mathrm{Im}(z^2)|} = \sqrt{2|xy|}, \quad v(x, y) = 0$$

$$\frac{\partial u}{\partial x}(0,0) = \lim_{\Delta x \to 0} \frac{u(\Delta x, 0) - u(0,0)}{\Delta x} = 0 = \lim_{\Delta y \to 0} \frac{v(0, \Delta y) - v(0,0)}{\Delta y} = \frac{\partial v}{\partial y}(0,0)$$

$$\frac{\partial u}{\partial y}(0,0) = \lim_{\Delta y \to 0} \frac{u(0, \Delta y) - u(0,0)}{\Delta y} = 0 = -\lim_{\Delta x \to 0} \frac{v(\Delta x, 0) - v(0,0)}{\Delta x} = -\frac{\partial v}{\partial x}(0,0)$$

f satisfies the Cauchy–Riemann equations at $z = 0$.

$$u(x, y) = \sqrt{|\mathrm{Im}(z^2)|} = \sqrt{2|xy|}, \quad v(x, y) = 0$$

$$\frac{\partial u}{\partial x}(0,0) = \lim_{\Delta x \to 0} \frac{u(\Delta x, 0) - u(0,0)}{\Delta x} = 0 = \lim_{\Delta y \to 0} \frac{v(0, \Delta y) - v(0,0)}{\Delta y} = \frac{\partial v}{\partial y}(0,0)$$

$$\frac{\partial u}{\partial y}(0,0) = \lim_{\Delta y \to 0} \frac{u(0, \Delta y) - u(0,0)}{\Delta y} = 0 = -\lim_{\Delta x \to 0} \frac{v(\Delta x, 0) - v(0,0)}{\Delta x} = -\frac{\partial v}{\partial x}(0,0)$$

Choose $\Delta z = \Delta x$

$$\lim_{\Delta z \to 0} \frac{f(\Delta z) - f(0)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\sqrt{2|\Delta x \cdot 0|} - \sqrt{2|0 \cdot 0|}}{\Delta z} = 0$$

Choose $\Delta z = \Delta x + i\Delta x, \quad \Delta x > 0$

$$\lim_{\Delta z \to 0} \frac{f(\Delta z) - f(0)}{\Delta z} = \lim_{\Delta x \to 0} \frac{\sqrt{2}\Delta x}{\Delta x + i\Delta x} = \frac{\sqrt{2}}{1 + i}$$

f is not differentiable at $z = 0$.

4. (a) Suppose f is analytic and real-valued in a domain D. Show that f is constant in D.

(b) Suppose f and $\overline{f}$ are analytic in a domain D. Show that f is constant in D.

(c) Suppose f is analytic in a domain D and $|f|$ is constant in D. Show that f is constant in D.

Solution

(a) $\operatorname{Im} f(z) = 0$ in $D \Longrightarrow \dfrac{\partial v}{\partial x} = \dfrac{\partial v}{\partial y} = 0 \Longrightarrow \dfrac{\partial u}{\partial x} = 0$

$\Longrightarrow f'(z) = \dfrac{\partial u}{\partial x} + i\dfrac{\partial v}{\partial x} = 0 \Longrightarrow f$ is constant in D.

(b) $\operatorname{Re} f(z) = \dfrac{1}{2}[f(z) + \overline{f(z)}]$ is real valued and analytic if both f and $\overline{f}$ are analytic. Hence $\operatorname{Re} f(z)$ is constant by (a). It follows that $f(z)$ is constant.

(c) **$|f(z)|$ constant in $D \Longrightarrow |f(z)|^2 = u^2 + v^2$ is constant in D. If $u = 0$ or $v = 0$ in D, then f is constant. Otherwise,**

$$\frac{\partial |f|^2}{\partial x} = 2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial |f|^2}{\partial y} = 2u\frac{\partial u}{\partial y} + 2v\frac{\partial v}{\partial y} = -2u\frac{\partial v}{\partial x} + 2v\frac{\partial u}{\partial x} = 0$$

$$\Longrightarrow \frac{1}{2}v\frac{\partial |f|^2}{\partial x} - \frac{1}{2}u\frac{\partial |f|^2}{\partial y} = 0 = (u^2 + v^2)\frac{\partial v}{\partial x}$$

$$\Longrightarrow \frac{\partial v}{\partial x} = 0 \Longrightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial u}{\partial y} = 0$$

$$\Longrightarrow f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = 0$$

$\Longrightarrow f$ **is constant in** D.

5. Show that the following functions are harmonic, and find its harmonic conjugates:

(a) $xy + 3x^2y - y^3$

(b) $e^{x^2-y^2}\cos(2xy)$

(c) $\tan^{-1}(y/x),\ x > 0$

(d) $\dfrac{x}{x^2 + y^2}$

Solution

(a) Set

$$u(x,y) = xy+3x^2y-y^3 \Rightarrow \begin{cases} u_x^{'} = y+6xy \\ u_y^{'} = x+3x^2-3y^2 \end{cases} \Rightarrow \begin{cases} u_{xx}^{''} = 6y \\ u_{yy}^{'} = -6y \end{cases} \Rightarrow \triangle u = 0,$$

thus is the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases} v_x^{'} = -u_y^{'} = -x-3x^2+3y^2 \Rightarrow v(x,y) = -\frac{x^2}{2} - x^3 + 3xy^2 + g(y) \Rightarrow v_y^{'} = 6xy + g'(y) \\ v_y^{'} = u_x^{'} = y+6xy \end{cases}$$

Thus we have that

$$g'(x) = y \Rightarrow g(y) = \frac{y^2}{2} + C,$$

thus

$$v(x,y) = \frac{y^2}{2} + 3xy^2 - \frac{x^2}{2} - x^3 + C.$$

In paranthesis we remark that the analytic function $f(z) = u + iv$ is given by

$$f(z) = xy+3x^2y-y^3+i\left(\frac{y^2}{2} + 3xy^2 - \frac{x^2}{2} - x^3 + C\right) = -iz^2(z+1/2)+iC.$$

(b) Set

$$u(x,y)=e^{x^2-y^2}\cos(2xy)\Rightarrow\begin{cases}u_x^{'}=2xe^{x^2-y^2}\cos(2xy)-2ye^{x^2-y^2}\sin(2xy)\\ u_y^{'}=-2ye^{x^2-y^2}\cos(2xy)-2xe^{x^2-y^2}\sin(2xy)\end{cases}\Rightarrow$$

$$\Rightarrow\begin{cases}u_{xx}^{''}=2e^{x^2-y^2}(\cos 2xy+2x^2\cos 2xy-2y^2\cos 2xy-4xy\sin 2xy)\\ u_{yy}^{'}=-2e^{x^2-y^2}(\cos 2xy+2x^2\cos 2xy-2y^2\cos 2xy-4xy\sin 2xy)\end{cases}\Rightarrow\triangle u=0,$$

thus the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases}v_x^{'}=-u_y^{'}=2ye^{x^2-y^2}\cos(2xy)+2xe^{x^2-y^2}\sin(2xy)\Rightarrow v(x,y)=e^{x^2-y^2}\sin(2xy)+g(y)\Rightarrow\\ \Rightarrow v_y^{'}=-2ye^{x^2-y^2}\sin(2xy)+2xe^{x^2-y^2}\cos(2xy)+g'(y)\\ v_y^{'}=u_x^{'}=2xe^{x^2-y^2}\cos(2xy)-2ye^{x^2-y^2}\sin(2xy)\end{cases}$$

Thus we have that

$$g'(y)=0\Rightarrow g(y)=C,$$

thus

$$v(x,y)=e^{x^2-y^2}\sin(2xy)+C$$

In paranthesis we remark that the analytic function $f(z)=u+iv$ is given by

$$f(z)=e^{x^2-y^2}\cos(2xy)+i\left(e^{x^2-y^2}\sin(2xy)+C\right)=e^{z^2}+iC.$$

(c) Set

$$u(x,y)=\tan^{-1}(y/x),x>0\Rightarrow$$
$$\Longrightarrow\begin{cases}u_x^{'}=\dfrac{1}{1+(y/x)^2}\left(-\dfrac{y}{x^2}\right)=\dfrac{-y}{x^2+y^2}\\ u_y^{'}=\dfrac{x}{x^2+y^2}\end{cases}\Longrightarrow$$
$$\Rightarrow\begin{cases}u_{xx}^{''}=\dfrac{2xy}{(x^2+y^2)^2}\\ u_{yy}^{'}=\dfrac{-2xy}{(x^2+y^2)^2}\end{cases}\Rightarrow\triangle u=0,$$

thus the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases}v_x^{'}=-u_y^{'}=\dfrac{-x}{x^2+y^2}\Rightarrow v(x,y)=a-\frac{1}{2}\log(x^2+y^2)+g(y)\Rightarrow v_y^{'}=\dfrac{-y}{x^2+y^2}+g'(y)\\ v_y^{'}=u_x^{'}=\dfrac{-y}{x^2+y^2}\end{cases}$$

Thus we have that

$$g'(y)=0\Rightarrow g(y)=C,$$

thus

$$v(x,y)=-\frac{1}{2}\log\left(x^2+y^2\right)+C.$$

(d) Set

$$u(x,y)=\frac{x}{x^2+y^2}\Rightarrow\begin{cases}u_x^{'}=-\dfrac{x^2-y^2}{(x^2+y^2)^2}\\ u_y^{'}=-\dfrac{2xy}{(x^2+y^2)^2}\end{cases}\Rightarrow\begin{cases}u_{xx}^{''}=-2x\dfrac{3y^2-x^2}{(x^2+y^2)^3}\\ u_{yy}^{'}=2x\dfrac{3y^2-x^2}{(x^2+y^2)^3}\end{cases}\Rightarrow\triangle u=0,$$

thus is the function $u(x,y)$ is harmonic. Cauchy Riemann give us

$$\begin{cases} v_x^{'} = -u_y^{'} = \dfrac{2xy}{\left(x^2+y^2\right)^2} \Rightarrow v\left(x,y\right) = \dfrac{-y}{\left(x^2+y^2\right)} + g\left(y\right) \Rightarrow v_y^{'} = -\dfrac{x^2-y^2}{\left(x^2+y^2\right)^2} + g^{'}\left(y\right) \\ v_y^{'} = u_x^{'} = -\dfrac{x^2-y^2}{\left(x^2+y^2\right)^2} \end{cases}$$

Thus we have that

$$g^{'}\left(y\right) = 0 \Rightarrow g\left(y\right) = C,$$

thus

$$v\left(x,y\right) = \frac{-y}{\left(x^2+y^2\right)} + C$$

In paranthesis we remark that the analytic function $f\left(z\right) = u + iv$ is given by

$$f\left(z\right) = \frac{x}{x^2+y^2} + i\left(\frac{-y}{\left(x^2+y^2\right)} + C\right) = \frac{\bar{z}}{\left|z\right|^2} + iC = \frac{\bar{z}}{z\bar{z}} + iC = \frac{1}{z} + iC.$$